

```
# Import required libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score

print("All libraries imported successfully!")
```

```
# Download and load boston housing dataset using kagglehub
import kagglehub
import os
```

```
path = kagglehub.dataset_download("vikrishnan/boston-house-
prices")
print("Path to dataset files:", path)
```

```
# Load dataset
df = pd.read_csv(os.path.join(path, "housing.csv"),
                 header=None,
                 sep='\s+',
```

```
names=['CRIM', 'ZN', 'INDUS', 'CHAS', 'NOX', 'RM', 'AGE',
'DIS', 'RAD', 'TAX', 'PTRATIO', 'B', 'LSTAT', 'PRICE'])
```

```
print("\nDataset loaded successfully!")
print(f"Shape: {df.shape}")
print(df.head())
```

```
# Check missing values
print("=== Missing Values ===")
print(df.isnull().sum())
```

```
print("\n=== Any Missing Values? ===")
print(df.isnull().values.any())
```

```
# Perform train test split
X = df.drop('price', axis=1)
y = df['price']
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=42)
```

```
# Create model variable and train the model
model = LinearRegression()
model.fit(X_train, y_train)
```

```
# Perform prediction and calculate error
y_pred = model.predict(X_test)
```

```
print(f"MSE: {mean_squared_error(y_test, y_pred):.2f}")
print(f"R2 Score: {r2_score(y_test, y_pred):.2f}")
```

```
# Plot Actual & Predicted price with linear regression line
plt.scatter(y_test, y_pred, alpha=0.7)
plt.plot([y.min(), y.max()], [y.min(), y.max()], color='red',
linestyle='--')
plt.xlabel("Actual Prices")
plt.ylabel("Predicted Prices")
plt.title("Actual vs Predicted Prices")
plt.show()
```

```
sample_data = X_test.iloc[0:1]
```

```
predicted_price = model.predict(sample_data)
```

```
print("Predicted House Price:", predicted_price[0])
print("Actual House Price:", y_test.iloc[0])
```